

# Performance Testing

**Date:** 15 March 2026    **Team ID:** 6ab4ccd16f290be8a6b460f4    **Project Name:** aiblognest

**Maximum Marks:** 5 Marks

Performance testing evaluates how the AI BlogNest API behaves under expected and peak load conditions. This document records the documented testing approach and provides a structured place for measured results.

## Step 1: Testing Overview

| Field               | Details                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Testing Tool Used   | Postman / Thunder Client collection is documented for API testing. A dedicated performance-load tool was not used. |
| Type of Testing     | Load Testing (planned/documented). Stress and spike testing are not evidenced in the supplied project materials.   |
| Target Module / API | AI BlogNest REST API: authentication, blog CRUD, AI blog generation, and AI summarization endpoints.               |
| Test Environment    | Local development environment; documented configuration uses port 5000.                                            |
| Test Date           | 15 March 2026                                                                                                      |

## Step 2: Test Scenarios

| S.No | Test Scenario / Description                                           | Virtual Users | Duration (sec) | Expected Outcome                                          |
|------|-----------------------------------------------------------------------|---------------|----------------|-----------------------------------------------------------|
| 1    | GET /api/blogs – retrieve all blogs under concurrent read load        | 10            | 60             | Responses remain within the target average response time. |
| 2    | POST /api/blogs – create blogs under concurrent write load            | 10            | 60             | Successful requests with stable response times.           |
| 3    | POST /api/ai/generate-blog – AI blog generation under concurrent load | 5             | 60             | AI requests complete successfully without errors.         |
| 4    | POST /api/ai/summarize – summarization under concurrent load          | 5             | 60             | Summarization requests complete successfully.             |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value          | Actual Value | Status  | Remarks                                             |
|------|----------------------|-----------------------|--------------|---------|-----------------------------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds           | Not measured | Pending | No measured load-test result was supplied.          |
| 2    | Response Time (Max)  | < 5 seconds           | Not measured | Pending | No measured load-test result was supplied.          |
| 3    | Throughput (Req/sec) | Record measured value | Not measured | Pending | Requires execution with a performance-testing tool. |
| 4    | Error Rate           | < 1%                  | Not measured | Pending | Requires measured request/error counts.             |
| 5    | CPU Utilization      | < 80%                 | Not measured | Pending | Requires monitoring during the load test.           |
| 6    | Memory Utilization   | < 80%                 | Not measured | Pending | Requires monitoring during the load test.           |

### Step 4: Observations & Analysis

The supplied project documentation confirms a Node.js/Express REST API with MongoDB/Mongoose, JWT authentication, blog CRUD operations, and Gemini-powered generation and summarization. The supplied Postman/Thunder Client collection documents functional API requests. However, no executed performance-test measurements, monitoring output, or screenshots were supplied. Therefore, the actual performance metrics are intentionally left as Pending rather than being estimated or fabricated.

**Recommended execution:** Run the four scenarios against the local API, record average/max latency, throughput, error rate, CPU utilization, and memory utilization, then replace the Pending values with the measured results and attach the tool screenshots.

### Step 5: Screenshots / Evidence

No performance-tool screenshots were supplied with the source material. Attach the JMeter, Locust, k6, or equivalent performance report/graphs here after execution.

#### Evidence checklist

- Load-test summary or report
- Response-time graph
- Throughput graph
- Error-rate result
- CPU and memory monitoring evidence